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ESTIMATION PRO — FIXED RATE BAND COST ESTIMATE

Project: Quick Analysis

Generated: 2026-05-26T15:47:08.157583+00:00

Model: STRUCTMIND CORE · PRO

SHA-256: 139ac690eab43df6868829fb5caca415fbba8850dcc24504ee61ee54c6ffa04f

CALCULATION MANIFEST — SINGLE SOURCE OF TRUTH

--- All locked input and output values for this estimate. Sections 1 and 7 copy from here.

Parameter	Value
AI-Extracted Fabricated Tonnage	14.68 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Material Factor	x1.00
Surface Factor	x1.00
Assembly Factor	x1.10
Composite Adjustment Factor	x1.10
Fixed Rate — Low (USD per ton)	$3,500
Fixed Rate — Mid (USD per ton)	$4,000
Fixed Rate — High (USD per ton)	$4,500
Adjusted Rate — Low	$3,850
Adjusted Rate — Mid	$4,400

Parameter	Value
Adjusted Rate — High	$4,950
Country	USA
Tax Rate Applied	0%
LOCK COST — LOW	$56,500
LOCK COST — MID	$64,600
LOCK COST — HIGH	$72,700

All LOCK values are fixed for this entire document. Sections 1 and 7 copy these values exactly. No re-derivation after this point.

1. EXECUTIVE SUMMARY

--- Project cost summary at fixed rate band.

Metric	Value
Project Name or ID	HOME 2 - CAROWINDS
Country	USA
Total Fabricated Tonnage	14.68 t
Material Type	Carbon Steel
Surface Treatment	Shop primer only
Assembly Complexity	Mixed bolt and weld
Composite Adjustment Factor	x1.10
Fixed Rate Band Applied	$3,500 — $4,500 per ton
Tax Rate	0%
Estimated Cost — Low (inc. tax)	$56,500

Metric	Value
Estimated Cost — Mid (inc. tax)	$64,600
Estimated Cost — High (inc. tax)	$72,700
Cost per Fabricated Ton — Mid	$4,400 per ton

Top Three Commercial Risks: 1. CRITICAL: Missing Lintel Scope: The quantity, sizes, and lengths of all W-beam and angle lintels (L1, L2, LLV types) are not provided and depend on unsupplied architectural drawings. This excluded scope could add 1-2 tons and a high piece count, impacting labor costs significantly. 2. WARNING: Assumed Member Lengths: All column lengths were estimated from floor elevations on sheets S2.03 and S2.04. No explicit stick lengths are provided in the schedules on sheet S0.02, creating a tonnage risk. 3. WARNING: Non-Standard & Unlisted Profiles: Profile HSS9x5x3/8 on sheet S1.02 is not a standard AISC section; mill availability and cost are unknown. Several other specified profiles are not in the standard AISC weight tables provided, requiring manual weight verification and potentially affecting material cost.

2. BASIS OF ESTIMATE

--- Drawings reviewed: S0.01 (General Notes), S0.02 (Schedules & Details), S1.01 (Foundation Plan), S1.02 (Second Level Framing Plan), S1.04 (Roof Framing Plan), S3.06 (Trellis and Site Wall Details).

Tonnage extraction method: (c) Estimated from framing plans where BOM is absent or incomplete. Member counts were taken from S1.01 and S1.02. Member lengths were estimated based on grid dimensions on S1.02 and floor elevations on S2.03/S2.04.

Composite factor selection: - Material (Carbon Steel): Explicitly specified on S0.01, Section 051200, Part 2.02 (ASTM A992 for W-shapes, A500 Gr. C for HSS, A36 for plates/angles). - Surface (Shop primer only): Selected based on the general requirement on S0.01, Section 051200, Part 2.05 A. Galvanizing requirement for bearing plates (S0.02) noted but not applied globally as it affects a minor tonnage percentage. - Assembly (Mixed bolt and weld): Inferred from typical connection details (1/S0.02 shows bolted connections) and standard fabrication practice for this building type. Beams will have shop-welded shear plates for field-bolted connections.

Rate basis: Fixed rate band of $3,500 (low) to $4,500 (high) per fabricated ton applied. Midpoint rate of $4,000 per fabricated ton used for the headline cost. These rates represent current USA structural steel shop fabrication pricing for the member types and complexity class identified in this package.

Tax applied: 0% for USA.

Global assumptions: - Column lengths are assumed to be 14'-0" (4,267 mm) based on elevations from S2.03. - The quantity of steel for the trellis detailed on S3.06 is estimated (10 columns, 5 beams) as no plan view was provided. - Profile HSS9x5x3/8 (S1.02) is assumed to be a typo; HSS10x4x3/8 was substituted for weight calculation. - All W-beam and angle lintels from schedules on S0.02 are EXCLUDED from this estimate due to missing quantities and lengths.

3. MEMBER-BY-MEMBER TONNAGE TAKE-OFF

--- Table 3A — Primary Structural Members Columns, primary beams, moment frames, transfer members, and lateral bracing.

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
1	Beam	—	W14x22	34	12'-10"	3,912	32.7	4,342	9,573	A992	S1.02	Plan Count
2	Beam	—	W16x36	2	22'-9"	6,934	53.6	743	1,638	A992	S1.02	Plan Count
3	Beam	—	HSS18x6x3/8	1	45'-1"	13,741	NOT IN TABLE	967	2,132	A500 Gr. C	S1.02	Plan Count
4	Beam	—	HSS18x6x3/8	2	29'-10"	9,093	NOT IN TABLE	1,280	2,822	A500 Gr. C	S1.02	Plan Count
5	Beam	—	HSS18x6x3/8	1	35'-10"	10,922	NOT IN TABLE	769	1,695	A500 Gr. C	S1.02	Plan Count
6	Column	C5	HSS6x6x3/8	14	Est. 14'-0"	4,267	33.1	1,979	4,363	A500 Gr. C	S1.01, S0.02	Estimated

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
7	Column	C6	HSS6x6x5/16	4	Est. 14'-0"	4,267	NOT IN TABLE	483	1,065	A500 Gr. C	S1.01, S0.02	Estimated
8	Column	C86	HSS8x6x3/8	2	Est. 14'-0"	4,267	NOT IN TABLE	399	880	A500 Gr. C	S1.01, S0.02	Estimated
9	Column	—	HSS8x6x3/8	10	Est. 10'-0"	3,048	NOT IN TABLE	1,427	3,146	A500 Gr. C	S3.06	Estimated
10	Beam	—	HSS8x6x3/8	5	Est. 13'-0"	3,962	NOT IN TABLE	927	2,044	A500 Gr. C	S3.06	Estimated

Table 3B — Secondary and Miscellaneous Members Secondary beams, purlins, girts, joists, bracing, stairs, handrails, and platforms.

No.	Type	Mark	Profile	Qty	Length (Imperial)	Length (mm)	Unit Wt (kg/m)	Est Weight (kg)	Est Weight (lb)	Grade	Source Sheet	Extraction Method
11	Header	—	HSS9x5x3/8	1	16'-11"	5,156	NOT IN TABLE	208	459	A500 Gr. C	S1.02	Plan Count
12	Angle	—	L6x4x1/4	1	16'-11"	5,156	NOT IN TABLE	57	126	A36	S1.02	Plan Count

Table 3C — Connection Material and Plates Shear plates, base plates, stiffeners, gusset plates, and embedded plates.

No.	Description	Thickness	Width	Length	Qty	Weight (kg)	Weight (lb)	Source Sheet	Extraction Method
13	C5 Base Plate	3/4"	12"	12"	14	194	428	S0.02	Schedule
14	C6 Base Plate	1"	12"	12"	4	74	163	S0.02	Schedule
15	C86 Base Plate	1"	16"	16"	2	66	146	S0.02	Schedule
16	C6 Cap Plate	1/2"	6"	6"	4	9	20	S0.02	Estimated
17	Trellis Base Plate	1"	16"	16"	10	329	725	S3.06	Detail

Table 3D — Tonnage Subtotals Consolidated weight by category feeding into the cost model.

Category	Count	Total Weight (kg)	Total Weight (lb)	Est. Tons
W-Shapes — Columns and Beams	36	5,085	11,211	5.09
HSS and Tube	34	8,438	18,595	8.44
Pipe	0	0	0	0.00
Angles and Channels	1	57	126	0.06
Plates (all types)	34	672	1,482	0.67
Miscellaneous and Embeds	0	0	0	0.00
Bolts and Accessories (3.00% addition)	—	428	943	0.43
TOTAL	105	14,680	32,357	14.68

4. COST BUILD-UP BY MEMBER CATEGORY

--- Per-category cost using the three-rate band. Adjusted rates = base rate x composite factor.

Category	Est. Tons	Adj. Rate Low (USD/ton)	Adj. Rate Mid (USD/ton)	Adj. Rate High (USD/ton)	Cost Low (USD)	Cost Mid (USD)	Cost High (USD)	Pct of Total
W-Shapes	5.09	$3,850	$4,400	$4,950	$19,597	$22,396	$25,196	34.7%
HSS and Tube	8.44	$3,850	$4,400	$4,950	$32,494	$37,136	$41,778	57.5%
Pipe	0.00	$3,850	$4,400	$4,950	$0	$0	$0	0.0%

Category	Est. Tons	Adj. Rate Low (USD/ton)	Adj. Rate Mid (USD/ton)	Adj. Rate High (USD/ton)	Cost Low (USD)	Cost Mid (USD)	Cost High (USD)	Pct of Total
Angles and Channels	0.06	$3,850	$4,400	$4,950	$231	$264	$297	0.4%
Plates	0.67	$3,850	$4,400	$4,950	$2,580	$2,948	$3,317	4.6%
Miscellaneous and Embeds	0.00	$3,850	$4,400	$4,950	$0	$0	$0	0.0%
Bolts and Accessories	0.43	$3,850	$4,400	$4,950	$1,656	$1,892	$2,129	2.9%
TOTAL	14.68	$3,850	$4,400	$4,950	$56,556	$64,636	$72,715	100%

5. PROCESS COST BREAKDOWN — MIDPOINT SCENARIO

--- Distributing $64,600 across industry-standard fabrication process activities. Percentages are fixed fabrication industry standards applied at midpoint.

Fabrication cost allocated across process activities at the midpoint scenario.

Process Activity	Industry Pct	Amount (USD)	Notes
Material — mill sections and plate	35%	$22,500	Structural steel mill cost
Shop labour — cut, fit, and weld	30%	$19,400	Includes AWS D1.1 weld operations
Coatings and surface preparation	12%	$7,800	SSPC class as specified
Consumables — weld wire, bolts, and gas	6%	$3,900	Per-ton allowance

Process Activity	Industry Pct	Amount (USD)	Notes
Equipment — plasma, burn table, drill	7%	$4,500	CNC and NC machine allocation
Overhead and QA/QC	6%	$3,900	Inspection, NDT, project management
Shop margin	4%	$2,600	Fabricator margin
TOTAL	100%	$64,600	Midpoint scenario

6. DRAWING COMPLETENESS AND COST EXPOSURE ASSESSMENT

--- Areas of drawing incompleteness that create commercial exposure for this estimate.

Area of Assessment	Status	Commercial Exposure if Unresolved
All quantities confirmed from BOM or member schedule	NOT CONFIRMED	HIGH: BOM is absent. All quantities estimated from plan views.
Material grades explicit on all members	CONFIRMED	LOW: Grades are specified in general notes on S0.01.
Surface treatment confirmed, not assumed	PARTIAL	MEDIUM: Base spec is primer, but galvanizing is noted for some items. Scope of galvanizing is unclear.
Assembly complexity confirmed from drawings	PARTIAL	LOW: Typical details are simple, but full connection design is not provided.
No conflicting BOM vs general notes grade callouts	NOT APPLICABLE	—

Area of Assessment	Status	Commercial Exposure if Unresolved
Drawings are IFC or internally coordinated	NOT CONFIRMED	MEDIUM: Discrepancies like non-standard HSS sections suggest coordination may be incomplete.
Galvanizing, AESS, or fire rating fully resolved	NOT CONFIRMED	MEDIUM: Galvanizing scope is ambiguous. No AESS or fire rating noted.

Items not confirmed create commercial exposure. The three fixed rates (low, mid, high) provide the price band buffer. No additional hidden contingency is applied.

Cost Waterfall — Midpoint Scenario Step-by-step cost build-up at the midpoint rate.

Step	Amount (USD)
Base cost — 14.68 t x $4,400/ton	$64,592
Tax at 0%	+$0
Grand Total — Low (at $3,500/ton base x composite x tax)	$56,500
Grand Total — Mid — Headline Cost	$64,600
Grand Total — High (at $4,500/ton base x composite x tax)	$72,700

7. COST CONVERSION SUMMARY

--- Final locked cost summary for this estimate.

Field	Value
Fabricated Tonnage	14.68 t
Fixed Rate Band Applied	$3,500 — $4,500 per ton
Composite Adjustment Factor	x1.10

Field	Value
Adjusted Rate — Low	$3,850 per ton
Adjusted Rate — Mid	$4,400 per ton
Adjusted Rate — High	$4,950 per ton
Country and Tax	USA at 0%
Grand Total — Low (inc. tax)	$56,500
Grand Total — Mid — Headline (inc. tax)	$64,600
Grand Total — High (inc. tax)	$72,700
Cost per Fabricated Ton — Mid	$4,400 per ton

8. ASSUMPTIONS AND EXCLUSIONS

--- Key Assumptions (minimum five specific items — cite sheet references): - Rate band of $3,500 to $4,500 per fabricated ton represents current USA market pricing for the member types, surface treatment, and assembly complexity identified in this estimate. - All steel lintels (W-shapes and angles per schedules on S0.02) are excluded from this estimate as quantities and lengths are missing from the provided drawing set. - All column lengths are assumed to be 14'-0" based on floor elevations shown on sheets S2.03 and S2.04. No schedule of lengths was provided. - The HSS9x5x3/8 profile specified on S1.02 is not a standard AISC section and has been substituted with HSS10x4x3/8 for weight calculation purposes. - The quantity of HSS columns and beams for the exterior trellis (detail 2/S3.06) has been estimated as 10 columns and 5 beams, as a plan view defining the full scope was not provided.

Exclusions — Items NOT included in this estimate: - Site erection labour, cranes, manlifts, rigging, and flagging. - Professional engineer stamping or delegated connection design. - Loose anchor bolts cast-in-place by the general contractor. - Site touch-up painting after final erection. - Tax or duties beyond 0% applied in this estimate. - 3D MEP coordination and cross-trade clash detection. - Inflation beyond 30 calendar days from the date of this estimate. - All steel lintels as noted in the Lintel Schedules on sheet S0.02. - Any steel required for the "Pre-fabricated canopy by others" noted on sheet S1.02.

Change Order Triggers: - Tonnage increases or decreases exceeding 5% from this estimate. - Coating system upgrade — for example, from primer-only to hot-dip galvanised for main members. - Connection type change from simple bolted to moment-resisting or heavily welded. - Addition of member categories not shown on the current drawing set, specifically the full scope of steel lintels. - Revision of drawings that changes profile sizes or eliminates members.

10. FINAL RECOMMENDATION

--- This estimate is preliminary and carries high commercial risk due to significant scope gaps in the provided drawings. The cost band should be used for budgeting purposes only, not for a fixed-price contract award. The most critical next action is to issue RFIs to clarify the full scope of steel lintels and confirm all member lengths. Re-estimation is mandatory once this information is provided, as the tonnage could increase by 10-15% and the piece count could double, materially impacting the final fabricated cost.

ACTION SUMMARY

--- Priority Actions for This Analysis

No.	Priority	Action Required	Owner	Deadline	Blocking
1	CRITICAL	RFI to Engineer/Architect: Request full schedule of all steel lintels (W-shapes and angles) with quantities, sizes, and lengths per S0.02 schedules.	Estimator	2 Days	Bid
2	HIGH	RFI to Engineer: Request confirmation of all column stick lengths. Assumed lengths based on elevations are not sufficient for fabrication.	Estimator	2 Days	Bid
3	HIGH	RFI to Engineer: Clarify non-standard profile HSS9x5x3/8 on S1.02. Confirm if typo and provide correct standard AISC section.	Estimator	3 Days	Material Purchase

No.	Priority	Action Required	Owner	Deadline	Blocking
4	MEDIUM	RFI to Engineer: Clarify the full scope of hot-dip galvanizing. Is it limited to bearing plates per S0.02 or does it include other exterior steel like trellis members?	Estimator	5 Days	Bid
5	MEDIUM	RFI to Engineer: Request plan view detailing the layout and full quantity of HSS members for the trellis shown on detail 2/S3.06.	Estimator	5 Days	Detailing